

Dhrumil Rana

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AI Systems Engineer building production AI, backend, and automation systems across full-stack surfaces | MS in Data Science, UT Arlington

EXPERIENCE

AI Systems Engineer | NovaChatAI

April 2025 to Present

- Joined as an early engineer at a fast-moving AI startup; **promoted from Intern to AI Systems Engineer within 3 months (July 2025)**. Own architecture and delivery across **8+ production surfaces** spanning conversational AI, messaging, PPV monetization, media, subscribers, campaigns, analytics, and authentication using **Python, FastAPI, PostgreSQL, Supabase, Next.js, and TypeScript**.
- Prevented duplicate and conflicting AI actions across conversation and monetization workflows using **persistent fan state, memory aware routing, execution guards, and database enforced uniqueness** across asynchronous automation.
- Built execution infrastructure for **mass messaging, campaign automation, scripts, and monetization**, supporting targeting, scheduling, delivery attempts, recovery, and execution history through relational state and indexed PostgreSQL access paths.
- Diagnosed a production AI outage with **9,000+ queued jobs** by tracing cron concurrency, worker saturation, queue state, and database records; restored processing, verified backlog recovery, and strengthened failure recovery controls.
- Cut interaction latency by **2–3 seconds across affected dashboard workflows** by eliminating dead clicks, moving blocking processes to background execution, and fixing UI stutter, while resolving **100+ production defects and functionality chokepoints**.

Software Developer Intern | Odoo

May 2024 to September 2024

- Developed **Python backend modules, REST APIs, and PostgreSQL business logic** for point of sale, inventory, and enterprise workflows while maintaining transactional correctness.
- Optimized **PostgreSQL queries and backend execution paths**, improving transaction performance by **15%** across production workflows.
- Resolved **50+ production defects** across application logic, databases, API integrations, and enterprise workflows while preserving existing business processes.

Research Analyst | Silent Info Tech Pvt. Ltd.

January 2024 to April 2024

- Built an end to end **Python ETL and analytics pipeline** processing **100K+ records across 20+ features**, producing analysis ready datasets for reporting and downstream modeling.
- Built **Pandas and NumPy data quality pipelines** to detect missing values, inconsistent formats, and invalid records before downstream analysis.
- Integrated **OpenAI language models** into content generation and transformation workflows, reducing measured manual processing effort by **80%**.

PROJECTS

Aegis Federate | Privacy Preserving Federated Intelligence

[GitHub](#)

- Built a **cross silo federated learning system** with Flower, gRPC, Docker, PyTorch, 1D CNNs, and MLPs for distributed training without centralizing raw data; implemented **DP SGD with Opacus** to evaluate privacy and predictive utility tradeoffs.

Agentic Cleanse | Autonomous Data Quality Engine

[GitHub](#)

- Built a **stateful multi agent ETL system** with LangGraph and Python for profiling, anomaly detection, schema reasoning, transformation, and validation, cutting manual review effort by **80% in benchmark testing** while retaining deterministic quality checks.

Twitter Scale Link Prediction Engine | Graph Machine Learning

[GitHub](#)

- Engineered a graph ML pipeline across **1.2M nodes with 1M+ DeepWalk sequences** and a 1,028 dimensional representation; trained a Random Forest model achieving **0.887 ROC AUC** over heuristic only link prediction approaches.

LEADERSHIP & RECOGNITION

Building Manager | UTA Maverick Activities Center

March 2025 to July 2026

- Led facility operations, emergency response, and coordination across a **70+ member workforce**; made time sensitive operational decisions and earned **Building Manager of the Year, 2025**.

Cultural Head | Hindu YUVA, University of Texas at Arlington

August 2025 to July 2026

- Led events for up to **1,000 guests** and taught performances to **150+ students**, coordinating choreography, volunteers, rehearsals, and event operations.

EDUCATION

Master of Science in Data Science | University of Texas at Arlington

January 2025 to July 2026

Bachelor of Engineering in Information Technology | Government Engineering College Modasa

November 2020 to May 2024

TECHNICAL SKILLS

- **AI and Machine Learning:** Generative AI, RAG, LangGraph, LangChain, PyTorch, TensorFlow, Scikit-learn, graph ML, federated learning, Flower, Opacus
- **Backend and Data:** Python, FastAPI, SQL, PostgreSQL, Supabase, REST APIs, schema design, indexing, migrations, background workers, schedulers, state machines, gRPC
- **Product and Analytics:** TypeScript, React, Next.js, Docker, Git, OAuth, RLS, Pandas, NumPy, ETL, Tableau, Power BI